



HOW TO USE THIS SET

20 questions, mixed difficulty, 6 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.

EASY

Which digestive enzyme is secreted almost entirely by the pancreas and stomach, hydrolysing proteins into polypeptides?

- A. Amylase
- B. Ribozymes
- C. Proteases
- D. Lipase
- E. Elastase alone

2.

EASY

What is the main catalytic effect of a digestive enzyme on a biochemical reaction?

- A. It lowers the activation energy of the reaction
- B. It acts non-specifically on many unrelated substrates
- C. It is consumed stoichiometrically in the reaction
- D. It changes the chemical nature of the end product
- E. It raises the activation energy to slow the reaction

3.

EASY

Which GI hormone contracts the gallbladder and stimulates pancreatic enzyme release in response to fat digestion products?

- A. Secretin
- B. Somatostatin
- C. Gastrin
- D. Cholecystokinin (CCK)
- E. Ghrelin

4.

EASY

Ghrelin, released by endocrine cells of the stomach in response to hunger, acts on which target to stimulate feeding?

- A. The duodenal mucosa
- B. The pancreas
- C. The liver
- D. The gallbladder
- E. The hypothalamus

5.

MEDIUM

CASE

A 26-year-old man has chronic diarrhoea and bloating after consuming dairy products. Stool studies show an acidic pH and osmotic gap.

Which mechanism best explains his symptoms?

- A. Deficiency of somatostatin increasing motility
- B. Overproduction of secretin inhibiting gastric emptying
- C. Increased pancreatic bicarbonate secretion
- D. Undigested lactose increases luminal osmotic pressure and undergoes bacterial fermentation
- E. Excess gastrin secretion increasing gastric acid

6.

MEDIUM

Endopeptidases differ from exopeptidases mainly in that endopeptidases:

- A. Require bile salts for activity
- B. Only cleave at the C-terminus
- C. Only cleave at the N-terminus
- D. Cleave internal peptide bonds within the protein chain
- E. Only act on carbohydrates

7.

MEDIUM

Trypsin is unique among the pancreatic proteases listed because, in addition to digesting protein, it also:

- A. Digests starch directly
- B. Is secreted by the salivary glands
- C. Requires pepsin for activation
- D. Activates other pancreatic zymogens such as chymotrypsinogen and procarboxypeptidase
- E. Neutralises gastric acid

8.

MEDIUM

Carboxypeptidase A and carboxypeptidase B are best distinguished from each other by:

- A. A acts in the stomach, B acts in the pancreas
- B. A digests starch, B digests protein
- C. A cleaves internal bonds, B cleaves external bonds
- D. A releases neutral/branched amino acids from the C-terminus, B releases basic amino acids (Arg, Lys) from the C-terminus
- E. A is an endopeptidase, B is an exopeptidase

9.

MEDIUM

CASE

A patient with chronic pancreatitis develops steatorrhoea, weight loss, and poor absorption of fat-soluble vitamins.

Which underlying mechanism best explains this presentation?

- A. Deficiency of pancreatic amylase, protease and lipase (exocrine pancreatic insufficiency)
- B. Lactase deficiency
- C. Excess gastrin secretion
- D. Somatostatin excess inhibiting bile flow
- E. Increased ghrelin secretion

10.

MEDIUM

Which statement correctly differentiates secretin and CCK in the duodenal phase of digestion?

- A. Secretin stimulates pancreatic bicarbonate; CCK stimulates pancreatic enzyme release and gallbladder contraction
- B. Secretin contracts the gallbladder; CCK stimulates bicarbonate secretion
- C. Secretin is released from the stomach; CCK is released from the pancreas
- D. Both stimulate only gastric acid secretion
- E. CCK inhibits pancreatic secretion entirely

11.

MEDIUM

Ribozymes are inactivated by which of the following?

- A. Binding to bile acids
- B. Heating past their melting temperature or altering essential nucleotides
- C. Cooling below body temperature
- D. Increasing luminal pH only
- E. Exposure to pepsin

12.

MEDIUM

A hormone that is classified among the 'local hormones' of the GI tract, controls GI secretion and motility, and is mostly peptide in nature — this description applies to:

- A. Catecholamines
- B. GIT hormones as a group (e.g. gastrin, CCK, secretin)
- C. Steroid hormones only
- D. Only pancreatic islet hormones
- E. Thyroid hormones

13.

MEDIUM

CASE

A researcher measures gastric acid output after infusing a peptide that is released by G cells in response to luminal amino acids.

Which hormone is being studied, and what is its principal action?

- A. Gastrin — increases gastric acid (HCl and pepsin) secretion and gastric motility
- B. Somatostatin — inhibits gastric acid secretion
- C. Ghrelin — stimulates feeding only
- D. GIP — inhibits gastric motility
- E. Secretin — inhibits gastric acid secretion

14.

MEDIUM

Which enzyme requires activation by hydrochloric acid or by autocatalysis from its own active form, rather than by another pancreatic enzyme?

- A. Elastase
- B. Carboxypeptidase B
- C. Chymotrypsin
- D. Trypsin
- E. Pepsin

15.

HARD

CASE

A patient with a somatostatinoma (somatostatin-secreting tumour) presents with symptoms of impaired digestion and altered glucose regulation.

Which combination of effects would be expected from excess somatostatin?

- A. Decreased insulin and glucagon secretion, decreased gastric acid/motility, and decreased pancreatic juice secretion
- B. No effect on pancreatic or gastric secretion
- C. Increased ghrelin with weight loss
- D. Increased gastrin and CCK secretion only
- E. Increased insulin and glucagon secretion with increased gastric acid

16.

HARD

A digestive enzyme deficiency panel shows normal lipase and protease but low amylase activity, with the patient otherwise asymptomatic for fat or protein malabsorption. Which enzyme source, active in tissues besides the pancreas and salivary gland, might partially compensate?

- A. Duodenal Brunner's glands secreting amylase
- B. Weak amylase activity present in liver, lung, fallopian tube, ovary and prostate
- C. Colonic bacteria producing amylase
- D. Hepatic bile acids substituting for amylase
- E. Gastric chief cells producing amylase

17.

HARD CASE

An investigator studies a patient in whom secretin release is chemically blocked after a fatty, acidic meal enters the duodenum.

Which downstream event would be most impaired as a direct consequence?

- A. Ribozyme activity in the gut would increase
- B. Gastrin secretion would rise sharply
- C. Ghrelin secretion would be suppressed
- D. Pancreatic bicarbonate secretion would fail to rise appropriately, and CCK's enzyme-stimulating effect would be blunted since secretin normally enhances it
- E. Gastric motility would increase

18.

HARD

Which pairing of a GI hormone with the specific pancreatic secretion it principally drives is correct, given that both are released in the duodenal phase but act on different pancreatic products?

- A. Gastrin → pancreatic bicarbonate; Somatostatin → pancreatic enzymes
- B. Secretin → pancreatic bicarbonate; CCK → pancreatic enzymes and gallbladder contraction
- C. Secretin → pancreatic enzymes; CCK → pancreatic bicarbonate
- D. GIP → pancreatic enzymes; Ghrelin → pancreatic bicarbonate
- E. CCK → gastric acid only; Secretin → gastric motility only

19.

HARD CASE

A 40-year-old woman with long-standing food intolerances, bloating, and undigested food in her stool is found to have reduced activity of several digestive enzymes attributed to chronic stress and repeated antibiotic use.

Which of the following is the most likely overall mechanism underlying her enzyme deficiency picture?

- A. Multifactorial acquired enzyme deficiency related to diet, antibiotic use, and stress rather than a discrete inherited defect
- B. A ribozyme-specific inactivation from dietary nucleotides
- C. A single-gene mutation causing zymogen misfolding
- D. Autoimmune destruction of gastric chief cells only
- E. Isolated pepsinogen deficiency from *H. pylori* infection

20.

HARD

Elastase's cleavage specificity for Ala, Gly and Ser residues, combined with its classification as an endopeptidase, means that on a mixed polypeptide substrate elastase would be expected to:

- A. Act exclusively on aromatic amino acid bonds
- B. Cleave internally wherever those small amino acids occur, generating multiple internal fragments rather than acting only at the chain ends
- C. Require bile acid activation before cleaving
- D. Only function in an alkaline duodenal environment above pH 9
- E. Trim only the terminal amino acid from each end of the chain

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	C	Proteases
2	A	It lowers the activation energy of the reaction
3	D	Cholecystokinin (CCK)
4	E	The hypothalamus
5	D	Undigested lactose increases luminal osmotic pressure and undergoes bacterial fermentation
6	D	Cleave internal peptide bonds within the protein chain
7	D	Activates other pancreatic zymogens such as chymotrypsinogen and procarboxypeptidase
8	D	A releases neutral/branched amino acids from the C-terminus, B releases basic amino acids (Arg, Lys) from the C-terminus
9	A	Deficiency of pancreatic amylase, protease and lipase (exocrine pancreatic insufficiency)
10	A	Secretin stimulates pancreatic bicarbonate; CCK stimulates pancreatic enzyme release and gallbladder contraction
11	B	Heating past their melting temperature or altering essential nucleotides
12	B	GIT hormones as a group (e.g. gastrin, CCK, secretin)
13	A	Gastrin — increases gastric acid (HCl and pepsin) secretion and gastric motility
14	E	Pepsin
15	A	Decreased insulin and glucagon secretion, decreased gastric acid/motility, and decreased pancreatic juice secretion
16	B	Weak amylase activity present in liver, lung, fallopian tube, ovary and prostate
17	D	Pancreatic bicarbonate secretion would fail to rise appropriately, and CCK's enzyme-stimulating effect would be blunted since secretin normally enhances it
18	B	Secretin → pancreatic bicarbonate; CCK → pancreatic enzymes and gallbladder contraction
19	A	Multifactorial acquired enzyme deficiency related to diet, antibiotic use, and stress rather than a discrete inherited defect
20	B	Cleave internally wherever those small amino acids occur, generating multiple internal fragments rather than acting only at the chain ends

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.